* 0000800000001 *

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CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

1615437621

INFORMATION AND COMMUNICATION TECHNOLOGY **0417/11**

Paper 1 Theory **May/June 2025**

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **12** pages. Any blank pages are indicated.

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1 Circle **two** output devices.

Actuator	Blu-ray disc	Cloud	Keyboard
Hard disk	Printer	SD card	Webcam

[2]

2 A computer system processes data by using applications software and system software.

(a) Describe the following types of applications software.

(i) Word processing

....

....

....

.... [2]

(ii) Control software

....

....

....

.... [2]

(iii) Apps

....

....

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.... [2]

(b) Data can be in two forms, analogue or digital.

Explain why analogue data needs to be converted to digital data.

....

.... [1]

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(c) Explain why digital data needs to be converted to analogue data.

....

.... [1]

3 Evaluate the use of smartphones for making video calls.

....

....

....

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....

.... [6]

4 Computer software needs to be tested with normal, extreme and abnormal data before it can be implemented.

(a) Explain what is meant by extreme test data.

....

.... [1]

DO NOT WRITE IN THIS MARGIN

(b) An input into the software is an integer between 1 and 10. The software uses a range check to test this value as it is entered.

State **one** piece of test data, from each of the following, that would be used to test the range check is working correctly.

(i) Normal test data

.... [1]

(ii) Extreme test data

.... [1]

(iii) Abnormal test data

.... [1]

5 Viewers can watch movies by video streaming. One of the issues with video streaming can be a reduction in quality due to file compression.

(a) Discuss the use of video streaming to watch movies.

....

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.... [6]

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(b) Explain what is meant by file compression.

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.... [3]

6 Many taxis now use satellite navigation systems.

Evaluate the use of satellite navigation systems for taxis.

....

....

....

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....

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....

.... [6]

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7 Explain, giving examples, the purpose of a corporate house style.

....

....

....

....

....

....

....

.... [4]

8 State the method of implementation from the following descriptions.

(a) The old system and the new system run together until the whole of the new system is implemented.

.... [1]

(b) The new system is implemented one branch at a time.

.... [1]

(c) The new system is implemented one process at a time.

.... [1]

9 A local area network (LAN) uses a hub.

(a) Explain the purpose of a hub.

....

....

....

....

....

.... [3]

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(b) Describe **one** drawback of using a hub on a local area network (LAN).

....

.... [1]

10 During the analysis of a current system, research needs to be carried out.

(a) Describe the characteristics of each of the following research methods.

(i) Observation

....

....

....

.... [2]

(ii) Interviews

....

....

....

.... [2]

(b) Following the analysis, design of the new system will take place.

Identify **four** items that need to be produced as part of the design of the new system

1 ....

2 ....

3 ....

4 .... [4]

DO NOT WRITE IN THIS MARGIN

(c) Following the implementation of the new system it needs to be tested.

Explain why the new system needs to be tested.

....

....

....

....

....

.... [3]

(d) A test plan includes expected outcomes of a test.

Write down **three** other elements found in a test plan.

1 ....

2 ....

3 .... [3]

11 A student is typing a report for his biology coursework and needs to check his spelling for errors. The student could use spell check software.

(a) Explain how spell check software works.

....

....

....

.... [2]

(b) Sometimes the spell check software highlights words or changes words which are correctly spelt.

Explain why this could have occurred.

....

....

....

.... [2]

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12 When sending emails users should follow netiquette.

(a) Explain what is meant by netiquette.

....

....

....

.... [2]

(b) One of the rules of netiquette is to avoid sending the same message repeatedly as this is considered to be spam.

Describe, giving an example, **three** other rules used in netiquette.

Rule 1..

....

Example 1..

....

Rule 2..

....

Example 2..

....

Rule 3..

....

Example 3..

.... [6]

13 **(a)** Describe what is meant by hacking.

....

....

....

.... [2]

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(b) One method of preventing a system being hacked is the use of passwords.

Describe the password rules that should be put in place to reduce the threat of hacking.

....

....

....

....

....

.... [3]

14 Give **one** example of a generic file extension that would be part of the following types of file.

(a) image

.... [1]

(b) text

.... [1]

(c) web page

.... [1]

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INFORMATION AND COMMUNICATION TECHNOLOGY **0417/21**

Paper 2 Document Production, Databases and Presentations **May/June 2025**

2 hours 15 minutes

You will need: Candidate source files (listed on page 2)

9547142316

INSTRUCTIONS

- Carry out **all** instructions in each step.
- Enter your name, centre number and candidate number on every printout before it is sent to the printer.
- Printouts with handwritten candidate details will **not** be marked.
- At the end of the examination, put all your printouts into the Assessment Record Folder.
- If you have produced rough copies of printouts, put a cross through each one to indicate that it is **not** the copy to be marked.
- You must **not** have access to either the internet or any email system during this examination.

INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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You have been supplied with the following source files:

j2521access.csv
j2521assembly.rtf
j2521class.csv
j2521evidence.rtf
j2521examinfo.rtf
j2521learners.csv
j2521mp_trends.csv
j2521phone.jpg
j2521results.rtf

Task 1 – Evidence Document

Open the file **j2521evidence.rtf**

Make sure that your name, centre number and candidate number will appear on every page of this document.

Save this document in your work area as **EVIDENCE** followed by your candidate number, for example, EVIDENCE9999

You will need your Evidence Document during the examination to place screenshots when required.

Task 2 – Document

You are going to edit a document about exams. The school uses a corporate house style for all its documents. Paragraph styles must be created and applied as instructed.

1 Using a suitable software package, open the file **j2521examinfo.rtf**

The page setup is set to A4, landscape orientation with 2.5-centimetre margins. Do **not** make any changes to these settings.

Three paragraph styles have already been created. Do **not** make any changes to these.

Save the document in your work area with the file name **ExamGuide**

Make sure it is saved in the format of the software you are using.

Place in your Evidence Document a screenshot to show this file has been saved. Make sure there is evidence of the file type.

[1]

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2 Place in the *ExamGuide* document:

- a centre-aligned header with automated page numbers
- a right-aligned footer on a single line with the text:
  Produced by: followed by a space then your name, centre number and candidate number.

Make sure that:

- all the alignments match the page margins
- no other text or placeholders are included in the header or footer areas
- headers and footers are displayed on all pages.

[3]

3 Select the subheading *General* and the following text up to and including the paragraph ending *… the evidence provided*.

Change the page layout so that only this text is displayed in two columns of equal width with a 2-centimetre space between the columns.

[2]

4 Apply a bulleted list to the text from

mobile phones …

to

… enabled electronic devices.

Format the bullets so that:

- the bullets are indented 2.5 centimetres from the left margin
- the list is in single line spacing with no space before or after each line
- there is a 6-point space after the last item in the list.

[3]

5 Locate the table in the document.

Delete the entire row and contents for the subject *Business Unit 1*

[1]

6 Sort all the data in the table from

Maths Paper 1

to

Combined Science Paper 3

into ascending order of *Exam Date* with data integrity maintained.

[1]

7 Apply to the table:

- 0.5-point gridlines to all cells
- 3- to 4-point external black borders

so that these are displayed when the table is printed.

[2]

8 Format the first row of the table so that:

- it becomes a single cell
- the text in the cell is centred vertically and horizontally
- it has a light grey (20–40%) background fill.

[3]

9 Apply the *EX-table* style to rows 2 to 12 only. [1]

10 Format the table so that:

- all text in each row displays on one line
- the table borders and all data fit within the column width
- there is a 6-point space after the table.

[2]

11 Create and store the following style, basing it on the default/normal paragraph style:

Style name	**Font style**	**Font size (points)**	**Alignment**	**Enhancement**	**Line spacing**	**Space before (points)**	**Space after (points)**
EX-subhead	sans-serif	18	centre	bold, underline	single	0	0

Take a screenshot to show that you have defined the settings for the *EX-subhead* style.

Make sure there is evidence that you have based this on the default/normal paragraph style.

Place this in your Evidence Document.

[2]

12 Identify the four subheadings in the document and apply the *EX-subhead* style to each one. [1]

13 In the last paragraph locate the text *Results Nomination Form*

Format this text so that when clicked it opens the document with the file name *j2521results.rtf*

Take screenshot evidence showing the text *Results Nomination Form* links to the correct file. Place this in your Evidence Document. Make sure that the file name is fully visible.

[2]

14 Spell check and proofread the document.

Make sure that:

- the list and table are **not** split over columns or pages
- there are no widows or orphans
- there are no blank pages
- the original styles are maintained
- spacing is consistent between all items.

Save the document using the same file name and format used in Step 1.

Print the document.

[1]

[Total: 25]

Task 3 – Database

You are now going to prepare a report. Dates must be imported in the format day month year (DMY). Make sure all required data is fully visible.

15 Use database software to import the file **j2521learners.csv** as a new table.

Use these field names and data types:

Field name	Data type	Display
Surname	Text	
Forename	Text	
Gender	Text	
Learner_no	Number	Integer
DOB	Date/Time	dd-MMM-yy *e.g. 12-Aug-09*
Telephone	Text	
School_email	Text	
Class_code	Text	
AA_code	Text	
Online_application	Boolean/Logical	To display as Yes/No

Set *Learner_no* as the primary key.

Save the data.

Place in your Evidence Document a screenshot showing the field names, data types and primary key used in the table.

[4]

16 Import the file **j2521access.csv** as a new table in your database. Set all data types to text.

Set *AA_code* as the primary key.

Place in your Evidence Document a screenshot showing the field names, data types and primary key used in the table.

[1]

17 Examine the file **j2521class.csv** and identify the most appropriate data to use as a primary key. Close this file.

Import the file **j2521class.csv** as a new table in your database.

Use these field names and data types:

Field name	Data type	Display
Year_group	Number	Integer
Class_tutor	Text	
Class_ref	Text	
Year_head	Text	
House_code	Text	
House_colour	Text	
House_leader	Text	

Select the most appropriate field from this data and use it to create a primary key.

Save the data.

Place in your Evidence Document a screenshot showing the field names, data types and primary key used in the table.

[2]

18 Create one-to-many relationships as links between:

- the *AA_code* field in the access table and the *AA_code* field in the learners' table
- the primary key field in the class table and the *Class_code* field in the learners' table.

Place in your Evidence Document screenshots showing the one-to-many relationships between the three tables.

[1]

19 Add the following as a new record in the learners' table:

Surname	**Norris**
Forename	**Emma**
Gender	**Female**
Learner_no	**10218427**
DOB	**10/07/2009**
Telephone	**0770090457**
School_email	**em_norris@tawara.ac**
Class_code	**4R**
AA_code	**AA02**
Online_application	**No**

[2]

20 Using fields from all the tables produce a tabular report that:

- selects the records where:
  – *Gender* is **Female**
  – *Access_arrangement* includes the text **time**
- shows only the fields *Learner_no*, *Forename*, *Surname*, *DOB*, *Gender*, *Year_group*, *Class_code*, *Access_arrangement* and *Online_application* in this order, with data and labels displayed in full. Do **not** group the data
- sorts the data into ascending order of *Access_arrangement* and descending order of *Year_group*
- has a page orientation of landscape
- fits a single page wide and prints on two pages only
- includes only the text **Access arrangements to be scheduled** as a title at the top of the page
- uses a function to display the *DOB* of the youngest student and place this at the end of the report
- has the label **DOB of youngest student** fully visible to the left of this date
- has your name, centre number and candidate number in the footer of the report so it appears in the same position on every page.

Save and print your report.

Place in your Evidence Document a screenshot showing the database formula used to find the youngest student. Make sure this formula is fully visible.

[10]

21 Using fields from all the tables produce labels which:

- select the records where:
  - *AA_code* is **AA01**
  - *Class_code* includes the numbers **5** or **6**
- are sorted in descending order of *Learner_no*

Use this selection to produce labels which:

- are arranged in two columns and four rows with eight labels to the page, for example, each label size 99.0 mm wide × 67.7 mm high (9.90 cm × 6.77 cm)
- print in portrait orientation with a page size of A4 (21 cm by 29.7 cm)
- display the data for each field left aligned with the layout as shown in the following sample label:

Access_arrangement *Learner_no* *Forename Surname* *DOB* *Class_code - Class_tutor*

- have your name, centre number and candidate number on the left at the bottom of each label.

Modify only the first two lines of the label so that the text:

- is displayed in a 14-point font size
- is centre aligned
- is displayed in bold.

Make sure that data on every label is fully visible with no overlap of any field.

Save and print your labels.

[10]

[Total: 30]

Task 4 – Printing the Evidence Document

Make sure that your name, centre number and candidate number appear on every page of your Evidence Document.

Save your Evidence Document.

Print your Evidence Document.

Task 5 – Presentation

You are going to create a short presentation.

All slides must have a consistent layout and formatting.

22 Create a presentation of 6 slides using the file **j2521assembly.rtf**

Unless otherwise instructed, the slides must display a title and a bulleted list.

[1]

23 Place in the centre of the slide header your name, centre number and candidate number.

Place in the footer automated slide numbers left aligned.

Make sure that:

- the header and footer appear in the same position on every slide
- no items overlap on any slide.

[2]

24 Format the first slide so that:

- a title layout is applied with no bullets
- the title and subtitle are centre aligned and in the middle of the slide.

[1]

25 Move the slide with the title ***Look after yourself*** so it becomes the last slide in the presentation. [1]

26 Import the image **j2521phone.jpg** and place it to the left of the bullets on the slide with the title *Prohibited items* [1]

27 Rotate the image 180 degrees. Make sure the aspect ratio is maintained. [1]

28 Use the data in the file **j2521mp_trends.csv** to create a vertical bar chart to show the number of offences for the category *Mobile phone use*. The chart must display only data from *2021* to *2024*. Display the years as labels on the category axis.

Do **not** display a legend.

[2]

29 Label the chart with the title **Increase in mobile phone offences** [1]

30 Display only the data values along the top of each bar. [1]

31 Format the value axis scale to display:

- a minimum value of **0**
- a maximum value of **2100**
- increments of **300**

[2]

32 Place the chart to the right of the bullets on the slide with the title *Tawara malpractice trends*

Make sure that:

- no words are split
- all data and labels are fully visible
- the chart and its contents do **not** overlap any slide items.

[1]

33 Save the presentation.

Print the full presentation in portrait orientation with 2 slides to the page, each filling half the page.
[1]

[Total: 15]

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INFORMATION AND COMMUNICATION TECHNOLOGY **0417/31**

Paper 3 Spreadsheets and Website Authoring **May/June 2025**

2 hours 15 minutes

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You will need: Candidate source files (listed on page 2)

INSTRUCTIONS

- Carry out **all** instructions in each step.
- Enter your name, centre number and candidate number on every printout before it is sent to the printer.
- Printouts with handwritten candidate details will **not** be marked.
- At the end of the examination, put all your printouts into the Assessment Record Folder.
- If you have produced rough copies of printouts, put a cross through each one to indicate that it is **not** the copy to be marked.
- You must **not** have access to either the internet or any email system during this examination.

INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **8** pages. Any blank pages are indicated.

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You have been supplied with the following source files:

TRbanner.png
TRdevelop.htm
TRracing.css
TRtest.csv
TRtext.txt
TRvideo.mp4

Task 1 – Evidence Document

Create a new word-processed document.

Make sure your name, centre number and candidate number will appear on every page of this document.

Save this Evidence Document in your work area as **j2531evidence_** followed by your
centre number_candidate number, for example j2531evidence_ZZ999_9999

You will need your Evidence Document during the examination to place screenshots when required.

Task 2 – File management

1 Create a new folder called **racing**

Locate only the following files and store them in your *racing* folder.

TRbanner.png
TRdevelop.htm
TRracing.css
TRtext.txt
TRvideo.mp4

Display the contents of your *racing* folder, showing the folder name and all file names, file extensions, file sizes, frame height, frame width and image dimensions.

Take a screenshot of this folder, making sure that the required information is clearly visible. Place this screenshot in your Evidence Document.

[1]

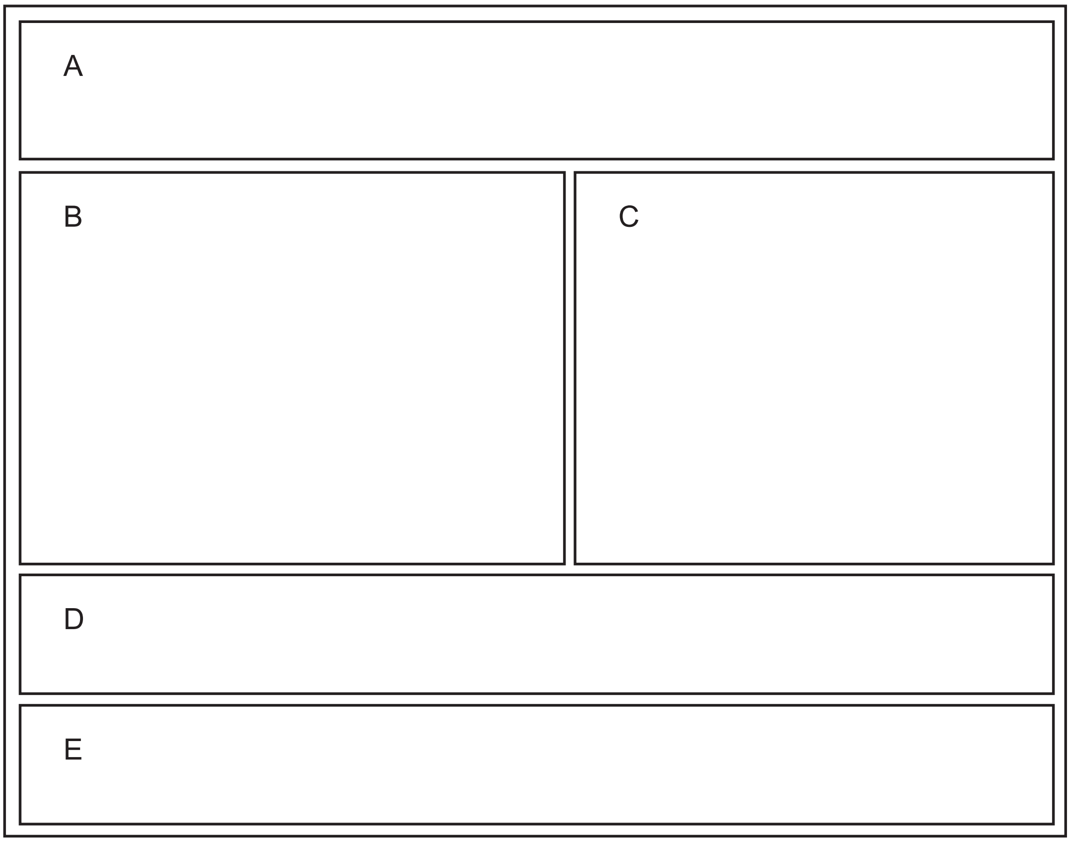
[Total: 1]

Task 3 – Web Page

You will create a web page to advertise Tawara Racing, who race motor cars.

2 Create in your *racing* folder a web page called **racing.htm**

This web page must be created using a single table and work in all browsers. The table must be left-aligned and fit 90% of the browser window. The table must have a structure as shown in this diagram:



Each table cell is identified with a letter which must **not** appear on your final web page.

Table borders must appear on the final web page.

[6]

3 Set the title of the web page to **Tawara Racing** [1]

4 Place in:

- cell A the image **TRbanner.png**
- cell B video and source tags to display **TRvideo.mp4**
  Make sure that the controls are visible and the video loops when played. Display an automated text-based error message if the browser does **not** support this video type.

[7]

5 Enter in cell C the text from the file **TRtext.txt**

Set this text as style h3.

[2]

6 Enter in cell D the text **Click here to find out more about our exciting developments, or here for job opportunities.**

Enter in cell E the text **Web page created by:** followed on a new line by your name, centre number and candidate number.

Set all the text in cells D and E as style h2.

[4]

7 Make the text **exciting developments** a link to open the web page **TRdevelop.htm** in a new window called **_blank**

Make the text **job opportunities** a link to an email editor to prepare an email to **Tawara.Racing@cambridge.org** with the subject line **Job opportunities**

[6]

8 Attach the stylesheet **TRracing.css** to your web page.

Edit this stylesheet so that the table borders are **not** collapsed.

Save this stylesheet.

[2]

9 Save your web page.

Take a copy of your HTML source and place this in your Evidence Document.

Display your web page in a browser. If necessary, resize it so that:

- the full page can be seen
- all text can be easily read
- the address bar is visible.

Take screenshot evidence showing your web page in the browser. Place this in your Evidence Document.

[1]

[Total: 29]

Task 4 – Printing the Evidence Document

Save and print your Evidence Document. Make sure your **name**, **centre number** and **candidate number** appear on every page of your Evidence Document.

Task 5 – Spreadsheet

You will assist Tawara Racing to analyse performance data when testing their racing cars.

10 Open and examine the file **TRtest.csv**

Place, right-aligned in the header, the text **Created by:** followed by a space then your name, centre number and candidate number.

Place in the footer the text **Created on:** followed by a space, the automated date, a space, then the text **at** followed by a space then the automated time.

Save this as a spreadsheet with the filename **TestData_** followed by your centre number_candidate number, for example TestData_ZZ999_9999

[3]

11 Enter the missing lap numbers in column A.

[1]

12 Place a replicable formula in cell F14 to calculate the time in seconds taken for this lap.

For example: if the minutes = 1, seconds = 14 and thousandths of a second = 484 this cell should display 74.484

1 minute = 60 seconds

Display the data in cell F14 to 3 decimal places.

Replicate this formula for all laps.

[6]

13 Insert ten rows between rows 1 and 2.

Move the contents of cells H13 to H20 into cells A3 to A10.

[2]

14 Merge cells A1 to F1.

Format this merged cell with a black 36-point sans-serif font on a yellow background.

Merge cells A12 to F12.

Format rows 1 to 13 of the spreadsheet to look like this:

	A	B	C	D	E	F
1	Test data for supercharger on the practice circuit					
2						
3	Fastest lap time:					
4	Slowest lap time:					
5	Average lap time:					
6	Laps with supercharger on:					
7	Laps with supercharger off:					
8	Supercharger on:					
9	Supercharger off:					
10	Result:					
11						
12	Lap times					
13	Lap	Supercharger	Minutes	Seconds	Thousandths of a second	Lap time
14	1	Off	1	14	484	74.484
15	2	On	1	12	675	72.675

[7]

15 Place in cell B3 a formula to display the fastest lap time from the data displayed. [2]

16 Place in cell B4 a formula to display the slowest lap time. [1]

17 Place in cell B5 a formula to display the average lap time. [1]

18 Place in cell B6 a formula to display the number of laps where the *Supercharger* was **On** [3]

19 Place in cell B7 a formula to display the number of laps where the *Supercharger* was **Off** [1]

20 Place in cell B8 a formula to calculate the average lap time when the *Supercharger* was **On** [4]

21 Place in cell B9 a formula to calculate the average lap time when the *Supercharger* was **Off** [2]

22 Place in cell B10 a formula to display **SC success** if the average lap times were faster with the supercharger on or to display **SC failed** if not.

Save your spreadsheet.

[4]

23 Print your spreadsheet showing the formulae. Make sure that the:

- printout is in landscape orientation
- row and column headings are displayed
- contents of all cells are fully visible.

[2]

24 Print your spreadsheet showing the values. Make sure that the:

- printout fits on a single page
- printout is in portrait orientation
- row and column headings are **not** displayed
- contents of all cells are fully visible.

[1]

[Total: 40]

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